

Curriculum Structure

Our comprehensive B.Tech program in Computational Engineering integrates mathematics, core engineering, scientific computing, and emerging technologies to prepare students for industry challenges and advanced research.

Program Overview

Courses related to Engineering Mathematics are included to lay the foundation of mathematics that a student of this program is expected to know. Core Engineering courses cover fundamental topics of Engineering Drawing, Thermodynamics, and Mechanics of solids and fluids along with the transport of heat and mass through these mediums. As biology forms an important aspect of modern engineering, courses on bioengineering and biomechanics have also been included. Scientific Computing lays foundation of the numerical methods typically used in engineering modelling and simulation; Data Structure & Analytics exposes the student to modern analysis tools of AI/ML and optimization; Computational Applications exposes the student to a vast array of methodology typically used in engineering analysis.

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Credit Distribution & Electives

Course Classification

CODE	CLASSIFICATION	CREDITS
BE	Basic Engineering	15
BS	Basic Science	13
DC	Departmental Core	61 (includes 3 from HPC Basket)
DE	Departmental Electives	17
LA/CA	Liberal/Creative Arts	9
FE	Free Electives	11
SK	Soft & Career Skills	3
Total Credits		129

Electives Offered

COURSE ID	COURSE NAME	CREDITS
CS5280	Concurrency Control in Transactional Systems	3
MS5973	Computational Thermodynamics & Kinetics	3
MS5033	Mesoscale Modelling	3
BT2023	Bio-Statistics	3
ME5810	Advanced CFD	3
ME5690	Advanced FEM	3
ME5470	Introduction to Parallel Scientific Computing	3
CH5190	Molecular Modelling of Catalytic Reactions	3
CS5300	Parallel and Concurrent Programming	3
CS5320	Distributed Computing	3
CO4015	Mini-Project	3

Parallel, Distributed & HPC Basket

ID	COURSE NAME	CREDITS
CS5280	Concurrency Control in Transactional Systems	3
ME5470	Introduction to Parallel Scientific Computing	3
CS5300	Parallel and Concurrent Programming	3
CS5320	Distributed Computing	3

Semester-wise Curriculum

Semester 1

COURSE ID	COURSE NAME	CREDITS	TYPE
MA1110	Calculus-1	1	BS
MA1220	Calculus-2	1	BS
EP1108	Modern Physics	2	BS
CY1010	Environmental Chemistry	2	BS
ID1063	Introduction to Programming	3	BE
ID1041	Engineering Drawing	2	BE
EE1202	Digital Circuits	3	DC
MS1980	Fundamentals of Scientific Computing	2	DC
BT1010	Introduction to Life Sciences	1	BS
Total Credits		17	

Semester 2

COURSE ID	COURSE NAME	CREDITS	TYPE
MA1140	Elementary Linear Algebra	1	BS
MA1150	Differential Equations	1	BS
ID1054	Digital Fabrication	2	BE
ME1020	Engineering Mechanics	3	BE
BM1030	Bioengineering	2	BE
CH1140	Thermodynamics Laws & Phase Transitions	3	DC
CO1020	Computer Aided Numerical Methods-I	2	DC
EE3402	Intro to Hardware Description Language	1	DC
EM3020	Introduction to Entrepreneurship	1	SK
	Personality Development	1	SK
Total Credits		17	